

Research Proposal

PhD Studentship: Adopting Agentic Artificial Intelligence for Building Supply Chain Resilience: A Multi-Layer Hybrid Predictive-Causal and Simulation-Based Framework

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Abstract

Global supply chains are increasingly exposed to systemic disruptions driven by geopolitical instability, environmental uncertainty, and operational complexity (Smyrnov et al., 2025). While Artificial Intelligence has significantly improved forecasting and optimisation, most implementations remain predictive, fragmented, and reactive, limiting their ability to support real-time resilience. This research proposes a novel, theory-driven framework that operationalises agentic Artificial Intelligence through a multi-layer architecture integrating predictive machine learning, causal configurational analysis, and simulation-based modelling. The study employs Extreme Gradient Boosting (XGBoost) for predictive analytics, Isolation Forest (iForest) for anomaly detection, reinforcement learning for adaptive decision-making, and Machine-Learning Enhanced Qualitative Comparative Analysis (mlQCA) for causal inference (Daramola et al., 2025; Pamisetty, 2024; Wu, et al., 2025). These components are systematically integrated within a unified data-model-decision pipeline, enabling continuous sensing, reasoning, and autonomous execution. Agent-based and stochastic simulation techniques are used to empirically validate resilience under diverse disruption scenarios. The research contributes to theory by extending dynamic capabilities into AI-driven, multi-agent ecosystems and reconceptualising resilience as a measurable and adaptive system property (Oyedepo, 2025). Methodologically, it introduces a hybrid predictive-causal framework that captures nonlinear interactions and multiple causal pathways. Practically, it provides an implementation-ready architecture for organisations to deploy resilient, explainable, and adaptive supply chain systems (Sharma and Joshi, 2024).

1.0 Background and Rationale

Modern supply chains operate as highly interconnected, multi-tier networks where disruptions propagate rapidly across global ecosystems. While interconnectivity enhances efficiency and scalability, it simultaneously amplifies systemic risk, making supply chains increasingly vulnerable to cascading failures triggered by geopolitical tensions, climate events, and operational disruptions. Traditional supply chain planning approaches, grounded in static forecasting and periodic optimisation, are insufficient in addressing the speed, complexity, and uncertainty of contemporary disruptions (Habibi et al., 2025; Itu, 2026).

Artificial Intelligence has improved supply chain operations through predictive analytics, enabling enhanced forecasting accuracy, risk detection, and operational efficiency. However, existing AI applications are predominantly reactive and functionally siloed, limiting their ability to support resilience as a dynamic and adaptive capability (Sharma and Joshi, 2024). These systems often lack integration across decision layers and fail to capture the nonlinear and configurational nature of disruption dynamics (Sharma and Joshi, 2024).

Recent advances in agentic Artificial Intelligence represent a paradigm shift. Agentic AI systems exhibit autonomy, reasoning, planning, and adaptive decision-making capabilities, enabling them to sense environmental changes, coordinate multi-agent actions, and execute decisions with minimal human intervention. These capabilities align closely with dynamic capabilities theory, which emphasises the ability of organisations to sense, seize, and reconfigure resources in response to environmental turbulence (Ajmal and Islam, 2026).

Despite these advancements, **significant gaps remain. Existing research lacks integrated frameworks that combine predictive intelligence, causal reasoning, and autonomous decision-making within a unified architecture.** Furthermore, resilience is often conceptualised rather than operationalised, limiting its measurability and practical applicability. **This research addresses these limitations by developing a hybrid, AI-native framework that integrates agentic intelligence with predictive modelling, causal analysis, and simulation-based validation.**

2.0 Research Aim and Objectives

Aim

To develop and empirically validate a theory-driven, hybrid predictive–causal and simulation-based framework that operationalises agentic Artificial Intelligence for enhancing supply chain resilience under conditions of uncertainty.

Objectives

1. To critically evaluate existing AI-driven supply chain resilience frameworks and identify theoretical and methodological limitations.
2. To develop a multi-layer conceptual framework integrating agentic AI capabilities across sensing, decision-making, and execution layers.
3. To design and implement a hybrid modelling architecture combining predictive machine learning, anomaly detection, reinforcement learning, and configurational causal analysis.
4. To develop and apply agent-based and stochastic simulation models to theoretically and computationally evaluate supply chain behaviour and resilience under diverse disruption scenarios.
5. To empirically identify causal configurations and boundary conditions under which agentic AI enhances resilience outcomes.
6. To develop an implementation roadmap for deploying agentic AI in real-world supply chain environments.

3.0 Research Questions

Primary Research Question:

How can agentic Artificial Intelligence be operationalised through hybrid predictive–causal and simulation-based models to enhance supply chain resilience under conditions of uncertainty?

Sub-questions:

1. What limitations exist in current AI-driven supply chain resilience frameworks?
2. How can agentic AI capabilities be structured across supply chain decision layers?
3. How do hybrid predictive–causal models improve resilience outcomes compared to traditional approaches?
4. How do reinforcement learning and simulation-based models influence disruption response and recovery dynamics?
5. What causal configurations of technological and organisational factors lead to high resilience performance?

4.0 Literature Review and Research Gap

4.1 Supply Chain Resilience

Supply chain resilience has evolved from a risk mitigation perspective to a dynamic capability encompassing anticipation, absorption, adaptation, and recovery. Contemporary frameworks emphasise time-based metrics such as time-to-recover and time-to-survive, alongside structural properties such as network redundancy and flexibility. However, many resilience models remain descriptive and lack integration with real-time decision systems (He et al., 2026; Sharma and Joshi, 2024).

4.2 Artificial Intelligence in Supply Chains

AI-driven supply chains leverage predictive analytics to improve forecasting, inventory management, and risk detection. While these systems enhance efficiency, they are often fragmented and lack cross-functional integration. Importantly, predictive models alone cannot capture disruption propagation or enable adaptive responses under uncertainty (Rejeb et al., 2025).

4.3 Agentic Artificial Intelligence

Agentic AI introduces autonomous, goal-directed decision-making capabilities, enabling systems to move beyond reactive analytics toward proactive orchestration. It supports multi-agent coordination, real-time adaptation, and

decentralised decision-making. Recent frameworks highlight its role in enabling autonomous supply chain ecosystems, yet empirical validation remains limited (Ranjan et al., 2025).

4.4 Hybrid Predictive Modelling

Hybrid modelling approaches combining statistical and deep learning methods improve forecasting robustness and adaptability in volatile environments. By integrating interpretability with nonlinear modelling, hybrid approaches address limitations of purely statistical or purely machine learning models (Fridkin and Bendersky 2026).

4.5 Simulation-Based Resilience Analysis

Agent-based modelling and stochastic simulation provide insights into disruption propagation and recovery dynamics. However, these approaches are often disconnected from predictive systems, limiting their applicability for real-time decision-making (Loganathan and Chinnaraju, 2025).

4.6 Causal Configurational Analysis

Traditional statistical approaches struggle to capture nonlinear and combinatorial relationships. Qualitative Comparative Analysis addresses this limitation by identifying causal configurations, but lacks integration with machine learning. **Emerging approaches such as mlQCA (fsQCA + XGBoost) combine predictive and causal reasoning, enabling more robust analysis** (Wu, et al., 2025).

4.7 Explainability and Governance

Explainability is critical for trust and adoption of AI systems. Agentic AI introduces additional complexity, requiring transparent and interpretable decision-making mechanisms. Research by Wu et al. (2026) emphasises the importance of AI literacy, trust calibration, and behavioural understanding in human–AI collaboration. Explainable AI techniques, such as SHAP ((SHapley Additive exPlanations)) values, enable stakeholders to understand model decisions, enhancing trust, accountability, and governance.

4.8 Research Gap

Despite significant advancements in Artificial Intelligence for supply chain resilience, **a critical gap remains in the absence of an integrated and empirically validated framework that unifies agentic AI capabilities with hybrid predictive modelling, configurational causal analysis, and dynamic simulation**. Existing studies predominantly focus on isolated predictive approaches or conceptual discussions of agentic systems, without systematically linking reinforcement learning, mlQCA, and simulation-based modelling. **As a result, current frameworks are limited in their ability to capture nonlinear interactions, threshold effects, and the complex, combinatorial conditions that drive resilience outcomes**. Furthermore, they fall short in enabling autonomous, adaptive decision-making required for effective performance under conditions of uncertainty. **This research addresses this gap by developing a unified, multi-layer framework that integrates prediction, causality, and simulation to operationalise resilience as a measurable and continuously adaptive capability** (Sharma and Joshi, 2024; Wu et al., 2026).

5.0 Research Methodology

This research adopts a multi-layer, AI-native architecture that explicitly operationalises agentic Artificial Intelligence across three interconnected layers:

- Sensing Layer (Perception and Data Acquisition)
- Decision-Making Layer (Prediction, Reasoning, and Intelligence)
- Execution Layer (Autonomous Action and Adaptation)

These layers are integrated within a continuous learning loop and validated through simulation.

5.1 Sensing Layer: Data Architecture and Environmental Perception

The sensing layer forms the foundation of the framework and is responsible for real-time data acquisition and situational awareness.

Scalable data pipelines will integrate structured and unstructured data from:

- Enterprise Resource Planning systems
- Internet of Things sensors
- Logistics and transportation systems
- External data sources such as weather, geopolitical indicators, and market signals

A hybrid cloud–edge architecture will support:

- Real-time data ingestion at the edge
- Large-scale processing and model training in the cloud

This layer enables continuous environmental scanning, aligning with the “sensing” capability in dynamic capabilities theory (Zungu and Laryea 2025).

5.2 Decision-Making Layer: Predictive, Causal, and Explainable Intelligence

The decision-making layer integrates predictive analytics, anomaly detection, and causal reasoning to generate actionable and interpretable insights for supply chain resilience.

Predictive and Anomaly Detection Models

Extreme Gradient Boosting (XGBoost) is employed as the primary predictive model for:

- Demand forecasting
- Supply chain risk prediction
- Logistics optimisation

XGBoost is selected due to its strong ability to model nonlinear relationships, handle high-dimensional and noisy data, and deliver robust generalisation performance (Wu et al., 2025).

To complement predictive capabilities, Isolation Forest is used for:

- Real-time anomaly detection
- Identification of emerging disruptions not captured in labelled datasets

This combination enables both forecasting of expected system behaviour and early detection of unexpected disruptions, forming a comprehensive predictive intelligence capability.

Causal Reasoning and Configurational Analysis

To move beyond prediction toward explanation and theory development, the study adopts Machine-Learning Enhanced Qualitative Comparative Analysis (mlQCA).

mlQCA integrates machine learning outputs with configurational causal analysis to:

- Identify causal pathways leading to high resilience outcomes
- Capture nonlinear interactions and threshold effects
- Reveal multiple equivalent configurations (equifinality)

This approach addresses the limitations of purely predictive models by enabling causal interpretation and theory-driven insight generation (Rejeb et al., 2025; Wu et al., 2025):

Explainability and Trust Mechanisms

Explainability is embedded across both predictive and causal components using SHAP values.

SHAP enables:

- Transparent interpretation of model predictions
- Quantification of feature contributions
- Consistent explanation across XGBoost and mlQCA outputs

This ensures that decision-makers can understand how and why specific outcomes are generated, supporting trust, accountability, and informed decision-making. The approach aligns with the emphasis on AI literacy and trust calibration in human–AI collaboration highlighted by Wu et al. (2025)

5.3 Execution Layer: Autonomous Decision-Making and Control

The execution layer operationalises decisions through adaptive and autonomous actions using reinforcement learning techniques, specifically Deep Q-Network (DQN) and Proximal Policy Optimisation (PPO) (Sharma and Garg, 2024)

Reinforcement Learning (DQN, PPO)

Applied for:

- Dynamic routing, inventory optimisation, and resource allocation in response to changing supply chain conditions
- Observing real-time environment states such as demand fluctuations, inventory levels, and network disruptions
- Selecting and executing optimal actions based on current system conditions
- Receiving feedback through reward or penalty signals aligned with performance objectives such as cost, service levels, and resilience
- Continuously updating and improving decision policies through iterative learning and adaptation

5.4 Cross-Layer Validation: Simulation Framework

This section integrates and validates all three layers.

Agent-Based Modelling

- Models supply chain nodes as autonomous agents
- Captures interactions, coordination, and disruption propagation

Monte Carlo Simulation

- Introduces uncertainty in demand, supply, and logistics
- Generates probabilistic outcome distributions

Scenario and Sensitivity Analysis

- Evaluates system performance under extreme disruptions
- Identifies critical vulnerabilities and resilience thresholds

This simulation environment ensures that:

- The sensing, decision-making, and execution layers function cohesively
- System behaviour is tested under realistic and extreme conditions

5.5 Evaluation Framework (Multi-Layer Performance Assessment)

Evaluation spans all layers:

Sensing Layer Metrics

- Data latency
- Data completeness and quality

Decision-Making Layer Metrics

- Predictive accuracy, precision, recall, AUC
- Causal consistency and coverage
- Explainability and trust indicators

Execution Layer Metrics

- Adaptive response velocity
- Autonomous decision effectiveness

Resilience Metrics (System-Level)

- Time-to-Recover (TTR)
- Time-to-Survive (TTS)
- Network redundancy and robustness

5.6 Empirical Validation Strategy

To ensure empirical validation of the proposed framework, the study will adopt a computational–empirical approach grounded in both real-world and high-fidelity simulated data environments. Where available, **secondary datasets from supply chain operations, logistics systems, and publicly accessible repositories will be utilised to train and evaluate predictive and anomaly detection models.** In addition, simulation environments will be parameterised using empirically observed distributions of demand variability, supplier reliability, lead-time fluctuations, and disruption patterns documented in prior literature and industry reports.

Agent-based and Monte Carlo simulations will generate large-scale synthetic datasets that reflect realistic supply chain dynamics under diverse disruption scenarios. These simulations will not function as purely theoretical constructs but as empirically informed experimental environments, enabling systematic testing of model behaviour, resilience outcomes, and decision effectiveness. Furthermore, benchmarking against traditional forecasting and rule-based systems will provide comparative empirical evidence of performance improvements.

Where feasible, the study will incorporate case-informed validation, drawing on industry-relevant scenarios to assess the practical applicability of the framework. **This multi-layered validation strategy ensures that findings are not only theoretically grounded but also empirically robust, generalisable, and relevant for real-world supply chain decision-making.**

5.7 Data Sources

This research adopts a multi-source data strategy combining real-world, secondary, and simulation-generated datasets to support both predictive modelling and empirical validation.

5.7.1 Secondary and Real-World Data

Where accessible, the study will utilise publicly available and industry-relevant datasets to reflect realistic supply chain conditions. These may include:

- Global logistics and transportation datasets
- Supply chain performance and disruption datasets
- Demand forecasting datasets (for example retail and FMCG demand data)
- Open datasets from platforms such as Kaggle, World Bank, and logistics repositories

These datasets will support:

- Demand forecasting
- Risk prediction
- Model benchmarking against real-world patterns

5.7.2 Enterprise and System-Generated Data

The framework is designed to integrate with enterprise systems such as:

- Enterprise Resource Planning systems
- Warehouse Management Systems
- Transportation Management Systems
- Internet of Things sensor data streams

Although direct access to proprietary datasets may be limited, the study will simulate these environments using structurally realistic data schemas to ensure applicability to real-world deployment contexts.

5.7.3 Simulation-Generated Data

To address limitations in data availability and to enable controlled experimentation, the study will generate synthetic datasets using:

- Agent-based modelling
- Monte Carlo simulation
- Scenario-based disruption modelling

Simulation parameters will be grounded in:

- Empirical distributions from literature
- Industry-reported disruption patterns
- Observed supply chain behaviours

This approach enables:

- Large-scale experimentation
- Testing under extreme and rare disruption scenarios
- Robust validation of resilience performance

5.7.4 Hybrid Data Integration Strategy

The study integrates real-world and simulated data within a unified analytical pipeline. Real-world datasets provide grounding and benchmarking, while simulation-generated data enables stress testing and exploration of complex, nonlinear interactions.

This hybrid approach ensures that the framework is:

- Empirically grounded
- Experimentally flexible
- Generalisable across diverse supply chain contexts

6.0 Plan of Work

Year 1

- Literature review and conceptual framework
- Development of research design
- Submission of Paper 1

Year 2

- Model development and integration
- Simulation experiments
- Submission of Paper 2

Year 3

- Empirical validation and case studies
- Framework refinement and implementation roadmap
- Submission of Paper 3

7.0 Significance and Contribution of the Research

This research makes theoretical, methodological, and practical contributions by advancing the integration of agentic Artificial Intelligence within supply chain resilience through a multi-layer, data-driven architecture.

7.1 Theoretical Contribution

This study extends existing literature by operationalising agentic Artificial Intelligence within supply chain resilience theory through a structured, multi-layer framework.

First, it advances dynamic capabilities theory by explicitly mapping:

- Sensing: data-driven environmental perception
- Decision-making: predictive and causal intelligence
- Execution: autonomous, reinforcement learning-based adaptation

This moves beyond firm-centric interpretations of dynamic capabilities toward AI-enabled, ecosystem-level coordination, aligning with emerging perspectives on decentralized and intelligent supply chain systems (Rejeb et al., 2025).

Second, the research contributes to resilience theory by reconceptualising resilience as a measurable, adaptive, and continuously optimised system property, rather than a static or post-disruption outcome. By embedding resilience metrics such as time-to-recover and disruption response velocity within an AI-driven architecture, the study provides a theoretically grounded and operational definition of resilience. of agentic (He et al., 2026).

7.2 Methodological Contribution

The study introduces a novel hybrid methodological framework that integrates predictive analytics, causal configurational analysis, and simulation within a unified system.

A key contribution is the explicit integration of data sources, analytical models, and evaluation metrics across layers, creating a coherent and closed-loop analytical pipeline:

- Predictive models (XGBoost, hybrid forecasting) transform real-world and simulated data into forecasts and risk signals
- Causal modelling (mlQCA with SHAP) converts predictive outputs into interpretable causal configurations, capturing nonlinear relationships and threshold effects
- Reinforcement learning models operationalise these insights into adaptive decision policies

Simulation frameworks (agent-based and Monte Carlo) validate system behaviour under diverse disruption scenarios

This integration addresses a major limitation in existing research, where predictive, causal, and simulation approaches are typically applied in isolation (Sharma and Joshi, 2024).

Furthermore, the use of Machine-Learning Enhanced Qualitative Comparative Analysis (mlQCA) represents a methodological advancement by combining:

- Scalability and predictive power of machine learning
- Configurational reasoning of QCA

This enables the identification of multiple causal pathways (equifinality) leading to resilience outcomes, which traditional statistical approaches fail to capture.

7.3 Practical Contribution

From a practical perspective, the research provides a deployable and scalable roadmap for organisations seeking to adopt agentic AI in supply chains.

The framework translates complex analytical processes into actionable decision-support capabilities, including:

- Real-time disruption detection
- Adaptive routing and inventory optimisation
- Dynamic supplier risk assessment
- Scenario-based resilience planning

Importantly, the integration of data sources, models, and decision layers ensures that the framework is not a conceptual abstraction but a system-ready architecture that can be embedded within enterprise environments such as Enterprise Resource Planning (ERP), Warehouse Management System (WMS), and Transportation Management System (TMS) platforms.

The inclusion of explainability through SHAP values ensures that decisions are transparent and interpretable, addressing critical concerns related to trust, governance, and adoption. This aligns with the emphasis on AI literacy and trust calibration highlighted by Wu et al. (2025).

Additionally, the framework supports multi-objective optimisation, enabling organisations to balance:

- Cost efficiency
- Service performance
- Sustainability goals

This is particularly relevant for decision-makers operating in volatile and resource-constrained environments.

7.4 Policy and Governance Contribution

The research contributes to policy and governance by embedding ethical, explainable, and human-in-the-loop AI principles within supply chain decision systems.

By integrating:

- Explainable AI (SHAP-based interpretability)
- Transparent decision pathways
- Human feedback loops

the study provides a foundation for responsible AI deployment in supply chains. This is critical in addressing regulatory, ethical, and societal concerns surrounding autonomous decision-making systems (Wu, et al., 2025).

7.5 Overall Contribution

Overall, this research delivers a first-of-its-kind integrated framework that:

- Combines predictive, causal, and simulation-based modelling
- Explicitly links data, models, and decision layers
- Operationalises agentic AI capabilities in real-world supply chain contexts
- Transforms resilience into a measurable, adaptive, and optimisable capability

By bridging the gap between theory, methodology, and application, the study provides both academic advancement and practical impacts.

8.0 Conclusion

This research proposes a novel and integrated approach to addressing supply chain resilience through agentic Artificial Intelligence. In an era characterised by persistent disruption and increasing supply chain complexity, the ability to move beyond reactive decision-making toward adaptive, intelligent, and autonomous systems is no longer optional but strategic. This research positions agentic Artificial Intelligence as a transformative capability that redefines how supply chains sense, decide, and act under uncertainty. By integrating predictive analytics, causal reasoning, and simulation within a unified multi-layer architecture, the study offers a rigorous and empirically grounded pathway for operationalising resilience as a continuous, data-driven capability. The proposed framework not only advances academic understanding of AI-enabled resilience but also provides a practical foundation for organisations seeking to design future-ready supply chains that are robust, adaptive, transparent, and sustainable. Through this integration, the research contributes to bridging the gap between conceptual innovation and real-world implementation, aligning technological advancement with organisational resilience and responsible AI governance.

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